

```
(.venv) kartikey@kartikey-LOQ-15IAX9:~/Kartikey/CS-216-Assignment-2$ python3
p2pkh_transactions.py
Wallet loaded
Address A: n1Kd21fkTz6osyLFghaEMsJAQHph7DaTUn
Address B: mmA4nz3qKS3vczYbH8p3voQYLwcfDVHpJt
Address C: mhZy6B9AT5tjUixshCetfD2kScurQWFm7P
Funding TX: 9ba4726b20d62e11928736420dbbabdf1ae0e9d9aa55990e6b6468a39e390169
UTXO A: 9ba4726b20d62e11928736420dbbabdf1ae0e9d9aa55990e6b6468a39e390169
Raw TX A->B:
020000000016901399ea368646b0e9955aad9e9e01adfabbb0d42368792112ed6206b72a49b010000
0000fdffffff016043993b000000001976a9143ddd71c62b7cbe8939888deb1624ec9e551a1cd588
ac00000000
Signed TX 1:
{
  'hex':
    '020000000016901399ea368646b0e9955aad9e9e01adfabbb0d42368792112ed6206b72a49b010000
    0006a47304402205a19228c7d2b9401712861ac95133b6fff817660276e6b00fb409c191919749e0
    2204c132a169827c48a28dc35ddd8a746a60061a638ba854c5e441ac093b956dd1c01210224c2280
    d52f7deaac1162967a10b2aa077b5b859145fea00c398f16d21bbdd11fdffffff016043993b000000
    001976a9143ddd71c62b7cbe8939888deb1624ec9e551a1cd588ac00000000',
    'complete': True
}
Decoded Transaction:
{
  'txid': 'ae52946f29830d749d769e0d3275d07e1ed568c5825272c87756a35b41f8ee46',
  'hash': 'ae52946f29830d749d769e0d3275d07e1ed568c5825272c87756a35b41f8ee46',
  'version': 2,
  'size': 191,
  'vsize': 191,
  'weight': 764,
  'locktime': 0,
  'vin':
    [
      {
        'txid':
          '9ba4726b20d62e11928736420dbbabdf1ae0e9d9aa55990e6b6468a39e390169',
        'vout': 1,
        'scriptSig':
          {
            'asm':
              '304402205a19228c7d2b9401712861ac95133b6fff817660276e6b00fb409c191919749e02204c1
              32a169827c48a28dc35ddd8a746a60061a638ba854c5e441ac093b956dd1c[ALL]
              0224c2280d52f7deaac1162967a10b2aa077b5b859145fea00c398f16d21bbdd11',
            'hex':
              '47304402205a19228c7d2b9401712861ac95133b6fff817660276e6b00fb409c191919749e02204
              c132a169827c48a28dc35ddd8a746a60061a638ba854c5e441ac093b956dd1c01210224c2280d52f
              7deaac1162967a10b2aa077b5b859145fea00c398f16d21bbdd11'
            },
            'sequence': 4294967293
          }
      ],
      'vout':
        [
          {
            'value': Decimal('9.99900000'),
            'n': 0,
            'scriptPubKey':
              {
                'asm': 'OP_DUP OP_HASH160
                3ddd71c62b7cbe8939888deb1624ec9e551a1cd5 OP_EQUALVERIFY OP_CHECKSIG',
                'desc': 'addr(mmA4nz3qKS3vczYbH8p3voQYLwcfDVHpJt)#7fttka78',
                'hex': '76a9143ddd71c62b7cbe8939888deb1624ec9e551a1cd588ac',
                'address': 'mmA4nz3qKS3vczYbH8p3voQYLwcfDVHpJt',
                'type': 'pubkeyhash'
              }
            }
          ]
        }
      ]
    }
  
```

```

    }
  }
]
}
Locking Script (ScriptPubKey):
{
  'asm': 'OP_DUP OP_HASH160 3ddd71c62b7cbe8939888deb1624ec9e551a1cd5
OP_EQUALVERIFY OP_CHECKSIG',
  'desc': 'addr(mmA4nz3qKS3vczYbH8p3voQYLwcfDVHpJt)#7fttka78',
  'hex': '76a9143ddd71c62b7cbe8939888deb1624ec9e551a1cd588ac',
  'address': 'mA4nz3qKS3vczYbH8p3voQYLwcfDVHpJt',
  'type': 'pubkeyhash'
}
TXID A->B: ae52946f29830d749d769e0d3275d07e1ed568c5825272c87756a35b41f8ee46
Raw TX B->C:
0200000000146eef8415ba35677c8725282c568d51e7ed075320d9e769d740d83296f9452ae000000
0000fdffffff01c0bc973b0000000001976a914168229caf26057d17f8f03545a9280a2432ca54088
ac000000000
Signed TX 2:
{
  'hex':
'0200000000146eef8415ba35677c8725282c568d51e7ed075320d9e769d740d83296f9452ae000000
0006a47304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a60
22079a00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b012102e32e378
06a0c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0fdffffff01c0bc973b000000
0001976a914168229caf26057d17f8f03545a9280a2432ca54088ac00000000',
  'complete': True
}
Decoded B->C:
{
  'txid': 'b088e29fcc359f92ad0b2b4fa6b77a325d8718ad768cb59ffbb16cc8843d1221',
  'hash': 'b088e29fcc359f92ad0b2b4fa6b77a325d8718ad768cb59ffbb16cc8843d1221',
  'version': 2,
  'size': 191,
  'vsize': 191,
  'weight': 764,
  'locktime': 0,
  'vin':
[
  {
    'txid':
'ae52946f29830d749d769e0d3275d07e1ed568c5825272c87756a35b41f8ee46',
    'vout': 0,
    'scriptSig':
{
      'asm':
'304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a6022079a
00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b[ALL]
02e32e37806a0c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0',
      'hex':
'47304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a602207
9a00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b012102e32e37806a0
c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0'
    },
    'sequence': 4294967293
  }
],
  'vout':
[
    {
      'value': Decimal('9.99800000'),
      'n': 0,
      'scriptPubKey':
{

```

```

        'asm': 'OP_DUP OP_HASH160
168229caf26057d17f8f03545a9280a2432ca540 OP_EQUALVERIFY OP_CHECKSIG',
        'desc': 'addr(mhZy6B9AT5tjUixshCetfD2kScurQWFm7P)#2zuxmgay',
        'hex': '76a914168229caf26057d17f8f03545a9280a2432ca54088ac',
        'address': 'mhZy6B9AT5tjUixshCetfD2kScurQWFm7P',
        'type': 'pubkeyhash'
    }
}
]
}
ScriptSig (Unlocking Script):
{
    'asm':
'304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a6022079a
00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b[ALL]
02e32e37806a0c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0',
    'hex':
'47304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a602207
9a00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b012102e32e37806a0
c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0'
}
TXID B->C: b088e29fcc359f92ad0b2b4fa6b77a325d8718ad768cb59ffbb16cc8843d1221

```

I tried using btcdeb (bitcoin debugger) to test the script is correct or not but could not find the correct syntax for it. I was running the below command.

```

./btcdeb
304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a6022079a0
0e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b01
02e32e37806a0c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0 OP_DUP
OP_HASH160 3ddd71c62b7cbe8939888deb1624ec9e551a1cd5 OP_EQUALVERIFY OP_CHECKSIG
but it kept giving the output as invalid script.
When I ran the same script on an online checker, it gave output as correct. A
video of the online checker has been attached to refer. The online website used
is https://wschae.github.io/build/editor.html.

```

UNDERSTANDING IT IN DETAIL

We take the example of the signed transaction 2. Below is the hex code of the transaction 2.

```

0200000000146eef8415ba35677c8725282c568d51e7ed075320d9e769d740d83296f9452ae000000
006a47304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a602
2079a00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b012102e32e3780
6a0c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0fdffffff01c0bc973b000000
001976a914168229caf26057d17f8f03545a9280a2432ca54088ac00000000
02000000 01 46eef8415ba35677c8725282c568d51e7ed075320d9e769d740d83296f9452ae
00000000 6a
47304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a6022079
a00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b012102e32e37806a0c
955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0 fdffffff
01c0bc973b000000001976a914168229caf26057d17f8f03545a9280a2432ca54088ac00000000

```

The first 4 bytes are the version number

Version: 02000000

The next byte is the input counter

Input Counter: 01

The next 32 bytes are the hash of tx_id of the previous transaction or the input

transaction being used
Hash of the spent transaction:
46eef8415ba35677c8725282c568d51e7ed075320d9e769d740d83296f9452ae

The next 4 bytes represent the index of the output in the output of previous transaction

TxOut index: 00000000 --> Means the first output of the previous transaction

The next byte represents the length of script

Script length: 6a --> $6 \times 16 + 10 = 106$ Bytes

The next 106 bytes represent the input script

Input Script:

47304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a6022079a00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b012102e32e37806a0c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0

Dissection of the input

47

304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a6022079a00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b01 21
02e32e37806a0c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0

Length of signature Data: 47 --> hex2dec --> 71 Bytes --> 142 Chars

<signature>:

304402201f151d22f3b879aa82b6967931b07efb6ccf5b4e3f753e8ac4facddcad7034a6022079a00e977a3e090829687d99d8d4e24a15b219262b110f13b651b28bf667938b01

Length of pubkey data: 21 --> hex2dec --> 33 Bytes --> 66 Chars

<pubkey>: 02e32e37806a0c955681dddae9b242ed13652d3cbecad19be90c0ca2308f2e7fc0

We use this signature and public key against the locking script of the transaction 1

The next 4 bytes represent the sequence

Sequence: ffffffff

The final bytes are the output

Output:

01c0bc973b000000001976a914168229caf26057d17f8f03545a9280a2432ca54088ac00000000
01 c0bc973b00000000 19 76a914168229caf26057d17f8f03545a9280a2432ca54088ac
00000000

The first byte is the number of outputs:

Number of outputs: 01

The next 8 bytes are the amount being sent

Amount: c0bc973b00000000 --> 3b97bcc0 --> hex2dec: 999800000 Satoshis

The next byte is the length of the output script:

Length: 19 --> hex2dec --> 25 Bytes --> 50 characters

The next 25 bytes are the output script:

Output script: 76 a9 14 168229caf26057d17f8f03545a9280a2432ca540 88 ac

76 --> OP_DUP

a9 --> OP_HASH160

Length: 14 --> hex2dec --> 20 Bytes --> 40 Digits

Pub key hash: 168229caf26057d17f8f03545a9280a2432ca540

88 --> OP_EQUALVERIFY

ac --> OP_CHECKSIG

00000000: Transaction locktime

We understand from this one complete hash of transaction B->C that it contains all the required information for the given transaction. It starts with the version number followed by the hash of the input being used. This hash is used to refer to the input transaction and hence the locking script that was associated with it.

The next bytes represent the input script or the unlocking script that contain the signature and the pubkey that are used against the locking script to prove that this transaction can use the previous UTXO.

Once the signature and pubkey are verified the transaction can take place. The previous UTXO is destroyed and new UTXO is created. Since there is no change left (i.e. all the amount of the previous UTXO is being sent) only one output exists.

A new locking script is created with this transaction that contains the hash of the public key of user C. When C wants to spend this UTXO, this locking script will be checked against the signature and the public key provided by the user C. To check that the UTXO being spent belongs to the spender, the locking script follows the below procedure.

`<signature> <pubkey> OP_DUP OP_HASH160 <pubkey_hash> OP_EQUALVERIFY OP_CHECKSIG`
This is a stack based algorithm. The user provides their signature and public key which are pushed into the stack.

The script duplicates the `<pubkey>` present at the top of the hash and pushes it into the hash.

The already present `<pubkey_hash>` in the script is then pushed into the stack. The `OP_EQUALVERIFY` function then pops out the top two elements and compares them. If they are equal then the script proceeds further otherwise stops.

The `OP_CHECKSIG` then pops out the `<pubkey>` and `<signature>` present in the stack and verifies that the signature belongs to the provided pubkey only. If true a 1 is pushed into the stack. This marks the end of the algorithm.

The full process is that we created a wallet named testwallet under which we created 3 legacy addresses (A, B and C). We first send 10 bitcoins to A (since we are in regtest we can directly send bitcoins using the function `sendtoaddress(<address>, <number_of_bitcoins>)`).

We then get the utxo set and find the utxo that is owned by A. This utxo has information of owner address, amount and the locking script which must be unlocked using the signature and the pubkey of the owner.

This utxo is then used as an input to send the bitcoins to B. A fee of 0.001 BTC is charged and a new locking script is created with the `pubkey_hash` of B. The previous UTXO is then destroyed and a new UTXO is created that has the address of B.

This new UTXO is then used in a new transaction of sending the bitcoins (now 9.999 BTC) to C with a fee charge of 0.001 BTC. The locking and unlocking are done again using the signature and pubkey of B and `pubkey_hash` of C respectively.

The new UTXO generated is now addressed to C and can be used for a new transaction when the signature and pubkey are provided.